

The ANNNI phase diagram under CNOT-target depolarizing noise

Benjamin Kaganovitch, Dimitrii Kashtanov, Yan Ivanov, Frederick Pu

Q-SITE Hacks 2026 • Quantum Coalition Scientific Open Challenge

Code, data and notebook: github.com/BenjaminKaganovitch/qsites-2026-open-challenge

Abstract. We map the ground-state phase diagram of the periodic ANNNI chain for $\kappa \in [0, 1]$, $h \in [0, 2]$ with sparse exact diagonalisation at $N = 8, 10, 12$ and, at $N = 8$, with a symmetry-preserving Hamiltonian-variational circuit (128 CNOTs) that reproduces the exact phase labels at 98.9% of grid points. We then evaluate the same circuit with a depolarizing channel on the target after every CNOT. At $p = 0.01$ the order parameters contract to about 60% of their clean values and the ferromagnet disappears under fixed thresholds, while boundaries defined by derivative peaks move by a median of only 0.005; at $p = 0.05$ the purity is close to that of the maximally mixed state. Channel-insertion sums show that the ferromagnet is the most fragile phase for this circuit. Richardson zero-noise extrapolation recovers 98% of the $p = 0.01$ labels; at $p = 0.05$ a two-point exponential extrapolation still recovers 97.5%, at a variance cost of about 5×10^4 . Every number below comes from the saved analysis outputs in the repository. AI coding assistants were used, as permitted by the organisers.

1 Problem and conventions

1.1 Model and diagnostics. We study the axial next-nearest-neighbour Ising (ANNNI) chain

$$H = -\sum_i Z_i Z_{i+1} + \kappa \sum_i Z_i Z_{i+2} - h \sum_i X_i, \quad (1)$$

with Pauli operators, $J_1 = 1$ and periodic boundary conditions, over $\kappa \in [0, 1]$ and $h \in [0, 2]$. Because the ground state of a finite symmetric ring keeps the global spin-flip symmetry, the one-point magnetisation $\langle Z_i \rangle$ vanishes even in the ordered phases [S07], so we identify the phases from longitudinal correlations instead. We use the structure factor

$$S(q) = \frac{1}{N} \sum_{i,j} e^{iq(i-j)} \langle Z_i Z_j \rangle, \quad (2)$$

with the self-terms kept, and define the floor-subtracted order parameters $O_F = (S(0) - 1)/(N - 1)$ and $O_A = (S(\pi/2) - 1)/(N/2 - 1)$, which equal 1 for perfect order and 0 for an uncorrelated state. The transverse magnetisation m_x completes the set. On ideal states the ferromagnet gives $S(0) = N$, the antiphase $S(\pi/2) = N/2$, and both $|+\rangle^{\otimes N}$ and $I/2^N$ give $S(q) = 1$, with m_x telling the last two apart.

1.2 Reference curves. The curves overlaid on the diagrams are approximate thermodynamic references rather than finite-ring labels [S10, S13]. They are the second-order Ising estimate (rationalised so that $h_I(0) = 1$), the lower antiphase-floating fit $h_{PT} = 1.05(\kappa - \frac{1}{2})$, which describes a Pokrovsky-Talapov transition [S12], and the upper Kosterlitz-Thouless (KT) fit $h_{KT} = 1.05\sqrt{(\kappa - \frac{1}{2})(\kappa - 0.1)}$, which the handout calls h_{BKT} . Our antiphase label absorbs the floating region between these two lines.

2 Methods

2.1 Exact reference. SciPy's sparse Lanczos solver returns the lowest six eigenpairs on 31×31 grids for $N = 8, 10, 12$, and on fine cuts of 21 values of κ by 401 values of h for all three sizes. Warm-started sweeps in h agree with independent cold starts to 10^{-13} . Degenerate ground spaces are treated as equal

mixtures and fidelities are measured between subspaces. The Hamiltonian matches the starter-kit exact diagonalisation and PennyLane's matrix representation to 7×10^{-15} .

2.2 Classifier. The thresholds are set from anchor regions fixed by limiting-case physics rather than from the overlay curves: the ferromagnet at $\kappa \leq 0.2$, $h \leq 0.2$; the antiphase at $\kappa \geq 0.8$, $h \leq 0.1$; and the paramagnet at $h \geq 1.8$. This gives $T = 0.541$ for the order parameters and $T_x = 0.490$ for m_x . A point is labelled ferro or antiphase if $\max(O_F, O_A) \geq T$, paramagnet if that fails but $m_x \geq T_x$, and unpolarised otherwise. Points whose label flips when the thresholds are moved by ± 0.1 are marked ambiguous. We also report a threshold-free relative-dominance label, $\arg \max(O_F, O_A, m_x^2)$. Under uniform terminal depolarisation all three quantities contract by the same factor a^{2m} with $a = 1 - 4p/3$ [S28], so in that benchmark this label is invariant under noise, and any change it does show reveals circuit-specific noise action.

2.3 Circuit. We use a Hamiltonian-variational ansatz (HVA) [S30]: the product state $|+\rangle^{\otimes N}$ followed by $L = N/2 = 4$ layers of

$$U_\ell = e^{-ib_\ell \sum_i X_i} e^{-ic_\ell \sum_i Z_i Z_{i+2}} e^{-ia_\ell \sum_i Z_i Z_{i+1}}, \quad (3)$$

which gives 12 parameters and preserves translation and parity symmetry. Each ZZ rotation compiles to CNOT-RZ-CNOT, so the circuit contains 128 CNOTs in total. A hardware-efficient RY/CNOT ring stalled at 1–3% energy error near the boundaries even at depth 6. Parameters are optimised at $p = 0$ by L-BFGS with exact adjoint gradients, warm-started sweeps and random restarts. Noise is PennyLane's `DepolarizingChannel(p)` applied to the target qubit after every CNOT on the `default.mixed` simulator; the 128 channels were verified on the circuit tape. A NumPy density-matrix engine that matches `default.mixed` to 3×10^{-14} runs the zero-noise-extrapolation and response sweeps.

2.4 Protocol. The main experiment uses the same parameters, circuit, grid, thresholds and colour scales at $p = 0, 0.01$ and 0.05 . The noise-induced shift $\delta h(\kappa, p) = h_b(p) - h_b(0)$ is measured for the same circuit, and the preparation bias $h_b(0) - h_b^{\text{ED}}$ is reported separately.

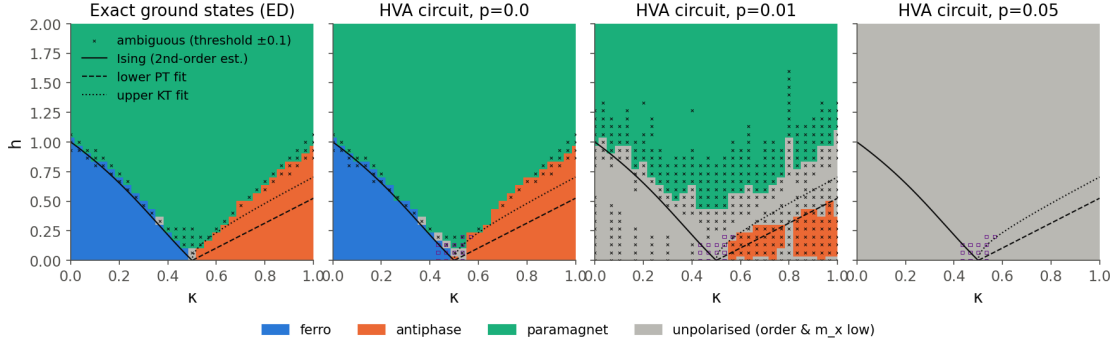


Figure 1. Phase diagrams on the 31×31 grid ($N = 8$, periodic). From left to right: exact ground states at $N = 8$ (the $N = 12$ map is in the repository’s finite-size figure), then the HVA circuit at $p = 0, 0.01$ and 0.05 with fixed parameters and fixed thresholds. Purple squares mark points where the clean preparation fidelity is below 0.9, and crosses mark threshold-ambiguous points. The curves are the Ising estimate (solid), the lower PT fit (dashed) and the upper KT fit (dotted).

3 Clean results and finite size

3.1 Circuit accuracy. At $p = 0$ the circuit is an accurate stand-in for the exact ground states. The median relative energy error is 4×10^{-5} and the median infidelity $1 - F$ is 5×10^{-5} , where F is the ground-subspace fidelity. Only 1.5% of grid points have $F < 0.9$, all of them around the multicritical point $\kappa \approx 0.5$, and the circuit labels agree with exact diagonalisation at 98.9% of points (Figure 1). The preparation bias of the threshold boundary is at most 0.034 outside $0.42 < \kappa < 0.6$ and reaches 0.066 at $\kappa = 0.57$, next to the multicritical point.

3.2 Finite-size behaviour. The boundary between the ferromagnet and the paramagnet is almost independent of system size and follows the Ising estimate. At $\kappa = 0.2$ the fidelity-susceptibility peak moves from 0.616 at $N = 8$ to 0.629 at $N = 12$, towards the DMRG value of 0.639 reported in [S10]; the second-order estimate is 0.653. The antiphase boundary, by contrast, drifts strongly with size. At $\kappa = 0.75$ the order crossing falls from 0.562 at $N = 8$ to 0.432 at $N = 12$, approaching the KT fit at 0.423 from above. The ring with $N = 10$ cannot hold a period-4 pattern and is visibly anomalous: at $\kappa = 0.6$ its crossing sits at 0.397, against 0.244 for $N = 8$ and 0.164 for $N = 12$. This is a commensurability effect of the finite ring, not physics of the thermodynamic limit.

3.3 Floating phase. At $N = 12$ the half-chain entropy of the pure ground state peaks close to the KT fit at $\kappa = 0.75$ and $\kappa = 0.9$, and the antiphase boundary drifts towards the KT line as N grows. Both are compatible with a critical region, but the dominant wavevector q^* stays locked at $\pi/2$ between the PT and KT fits at every size. The only jumps in q^* occur above the KT line, inside the modulated paramagnet [S10], and the momentum resolution of $\Delta q = \pi/6$ at $N = 12$ cannot resolve an incommensurate modulation. We therefore do not claim a floating phase [S12, S23]. A label-free k -means clustering on $(S(q)/N, m_x)$ reproduces the clean rule labels at 99.1% of points (a consistency check, not an independent validation).

4 Noise: effects, magnitude and mechanism

4.1 Noise level $p = 0.01$. The order parameters contract to a median of 0.60 of their clean values, compared with 0.65 if every target hit acted as end-of-circuit noise. The contraction depends on the phase. In the limit $h \rightarrow 0$, O_F at $\kappa = 0.2$ drops from 1.00 to 0.37, whereas O_A at $\kappa = 0.9$ drops only to 0.59. Under the fixed threshold the ferromagnet disappears entirely: none of the points that exact diagonalisation labels ferro keep their label, while 46% of antiphase points and 89% of paramagnet points survive. This is a loss of absolute signal rather than of the phase itself: re-applying our anchor rule at $p = 0.01$ gives thresholds $T = 0.266$, $T_x = 0.372$, recovers all 142 ferro points and agrees with exact diagonalisation at 96.1% of points. Wherever a threshold crossing does survive it moves to lower h , by 0.04 to 0.48. This has the same sign as the exact first-order prediction $-p \partial_p O / \partial_h O$ but is 1.4 to 3.8 times larger, because the response is already nonlinear at this noise level. The boundaries defined by derivative peaks, on the other hand, barely move (Table 1): the median $|\delta h|$ is 0.005, below the seed-to-seed spread of 0.011 and therefore indistinguishable from zero, and 28 of the 30 defined rows stay within 0.03, which is half a grid step. The exceptions are $\kappa = 0.13$ and 0.27 , which shift by $+0.04$, and $\kappa = 0.93$, which shifts by -0.12 . This is the behaviour that [S28] predicts for a near-uniform rescaling of the observables, and accordingly the relative-dominance map at $p = 0.01$ agrees with the clean exact-diagonalisation map at 97% of points, compared with 99% at $p = 0$.

4.2 Noise level $p = 0.05$. At this noise level the median purity is 0.0047, against 0.0039 for $I/2^8$, and every point is unpolarised under the fixed thresholds. Purity, however, says little about local observables: m_x keeps 16% of its clean value, so m_x^2 keeps about 2.6%, while the order parameters keep a median of 8.5%. In the relative-dominance map it is therefore the paramagnet that shrinks, and at large κ the deep paramagnet reads as antiphase. Re-applying the anchor rule at $p = 0.05$ gives $T = 0.026$, $T_x = 0.094$ and 69% agreement with exact

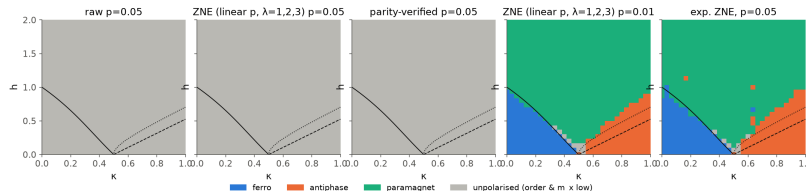


Figure 2. Error mitigation with fixed thresholds. From left to right: the raw map at $p = 0.05$, Richardson extrapolation at $p = 0.05$, parity verification at $p = 0.05$, Richardson extrapolation at $p = 0.01$ ($\lambda = 1, 2, 3$), and two-point exponential extrapolation at $p = 0.05$ ($\lambda = 1, 2$). Richardson restores the $p = 0.01$ diagram; only the exponential model restores $p = 0.05$.

Table 1. Boundary positions and noise-induced shifts for the HVA circuit. h_T is the fixed-threshold crossing and h_D the position of the maximum of $-dO/dh$. The grid step is 0.067; the seed-to-seed spread of δh_D at $p = 0.01$ has median 0.011 and maximum 0.038. The full table for all 31 values of κ is in the repository.

κ	Order	h_T^{ED}	Prep. bias	$h_D(0)$	$\delta h_D(p=0.01)$	$\delta h_D(p=0.05)$	h_T at $p=0.01$
0.2	ferro	0.685	-0.000	0.653	+0.027	+0.64 [†]	absent
0.3	ferro	0.487	-0.016	0.469	+0.015	+0.19	absent
0.4	ferro	0.246	+0.000	0.248	+0.003	+0.006	absent
0.7	anti	0.467	+0.018	0.477	-0.001	-0.004	absent
0.8	anti	0.650	+0.024	0.672	+0.003	+0.045	absent
0.9	anti	0.809	+0.004	0.723	+0.000	+0.098	absent

[†] Unreliable: the order signal at $p = 0.05$ is too small to locate a derivative peak.

diagonalisation, so a weak but real signal remains.

4.3 Mechanism from exact channel-insertion sums. We evaluated the first-order response $\partial_p \langle O \rangle = \sum_g \text{Tr}\{O_g \mathcal{L}_t(\rho_g)\}$ one channel location at a time [S28]. For the ferromagnetic state $d \ln O_F / dp = -99$, for the antiphase $d \ln O_A / dp = -69$, and for m_x in the paramagnet the value lies between -32 and -48. The values that end-of-circuit noise would give are -43 for pair correlators and -21 for m_x , so the measured sensitivities exceed their terminal baselines by factors of 2.3 (ferro), 1.6 (antiphase) and 1.5 to 2.3 (m_x). Errors that occur mid-circuit are carried through the later CNOT-RZ-CNOT gadgets and damage the ZZ order more than errors at the end of the circuit would. The early layers contribute the most: for the ferro point, dO_F / dp receives -27 from layer 1 and -17 from layer 4. The difference between the phases is not a property of the states themselves: a single flipped qubit removes the same fraction, about $4/N$, of the correlators that enter O_F or O_A , and the terminal baseline is identical for both. The gap between -99 and -69 comes from how mid-circuit errors propagate through the gadgets, together with the fact that the ferromagnetic state needs the full depth (Section 4.4). This ranking is specific to this circuit and to these diagnostics, and it reverses the organiser’s hypothesis [S01]. The noise does not change the spectrum of the Hamiltonian; what these maps show is the degradation of the prepared state.

4.4 Depth trade-off. We repeated the fixed-parameter experiment with $L = 1$ to 4 layers on four cuts in κ . The clean error $\langle |O - O^{\text{ED}}| \rangle_h$ falls with depth while the noise error grows with it, so the best depth shrinks as p grows. The antiphase is already accurate at $L = 2$, with a clean error of 0.024 at $\kappa = 0.8$, and $L = 2$ remains the best depth there at both $p = 0.01$ and $p = 0.05$. The ferromagnet needs $L = 4$ to be accurate, since its error at $L = 2$ is 0.17, and at $p = 0.01$ its best depth is $L = 3$. The ferromagnetic state therefore needs the full depth and carries the most noise channels.

5 Error mitigation

5.1 Zero-noise extrapolation. We ran Richardson extrapolation on every correlator C_{ij} and every $\langle X_i \rangle$ evaluated at λp for $\lambda = 1, 2, 3$ (linear noise scaling), holding out the $p = 0$ truth, and rebuilt $S(q)$ before classifying. At $p = 0.01$ this is effective (Figure 2): the median $|\Delta C_{ij}|$ falls from 0.063 to 0.011, $|\Delta m_x|$ falls from 0.26 to 0.023, and label agreement with the clean circuit rises from 70% to 98%. The threshold boundaries return to within $|\delta h| \leq 0.09$ with a median of 0.03; at $\kappa = 0.93$, for example, the shift goes from -0.48 to -0.04. The price is a variance-amplification factor of $\sum \gamma^2 = 19$. At $p = 0.05$ Richardson extrapolation fails for both linear and composed-channel scaling, because the correlators decay nearly exponentially in p ($d \ln O / dp \approx -70$ to -100) and a polynomial through $\lambda = 1, 2, 3$ cannot follow that. A two-point exponential model, $C_0 = C_1^2 / C_2$ from $\lambda = 1, 2$, does: at $p = 0.05$ it brings label

agreement with the clean circuit from 0.6% to 97.5% (96.4% with exact diagonalisation), with median $|\Delta C_{ij}| = 0.062$ and $|\Delta m_x| = 0.089$, and it restores the threshold boundaries at $\kappa = 0.2, 0.3, 0.8$ and 0.9 to within -0.095, -0.007, +0.001 and +0.026 of the clean circuit. The information therefore survives at $p = 0.05$ in exact simulation, but the median variance amplification of 5.5×10^4 means that recovering it on hardware would cost that many times more shots. At $p = 0.01$ the same model reaches 99.5% agreement with a factor of 18.5.

5.2 Parity verification. Post-selecting on the parity operator $P = \prod_i X_i$ accepts 71% of the weight at $p = 0.01$ and 51% at $p = 0.05$. It lowers the error in C_{ij} only from 0.063 to 0.053, because most errors stay inside the even-parity sector.

5.3 Noise-aware re-optimisation. On the cuts $\kappa = 0.2, 0.3, 0.7$ and 0.8 (8 values of h each) we minimised $\text{Tr}(H\rho_p)$ with exact gradients through the channels. The noisy energy falls by 0.2 to 1.0 per cut, but the order parameters move by only -0.016 to +0.028 and their error against exact diagonalisation is unchanged (0.225 to 0.198 at $\kappa = 0.8, p = 0.01$). The optimiser gains energy through m_x (+0.03 to +0.09) by leaving the ground state: at $\kappa = 0.2, h = 0$ the clean energy of the re-optimised parameters is -5.83 against -6.40. Energy-optimal noisy states are not diagnostic-optimal.

6 Conclusions and limitations

The clean phase diagram is reproduced accurately on rings of 8 to 12 sites. For a symmetric circuit accurate enough to represent these ground states, which costs 128 CNOTs, noise at $p = 0.01$ already removes absolute ferromagnetic order under fixed thresholds while the boundary positions defined by derivatives stay put, and $p = 0.05$ leaves only a weak signal that polynomial extrapolation cannot use. Richardson extrapolation recovers the $p = 0.01$ map almost fully, and an exponential model recovers the $p = 0.05$ map at a very large variance cost. Because the floating phase is unresolved at $p = 0$, we make no claim about it at $p > 0$; that region is uniformly unpolarised at $p = 0.05$. A Trotterised quench from $|+\rangle^{\otimes N}$ shows a sharp Loschmidt-rate peak only for quenches deep into the ordered side ($h \leq 0.4$), and at $p = 0.01$ the channel error is about eight times the Trotter error, so the dynamical signature does not survive noise. The limitations are the small system sizes, the approximate overlay curves, the masking of the $\kappa \approx 0.5$ region for preparation quality, the use of the fixed-parameter protocol on the full grid with re-optimisation only on cuts, and the absence of shot noise in these exact simulations.

References. Full entries are in the repository’s literature notes. [S01] Organiser handout; [S07] starter-kit audit; [S10] Beccaria, Campostrini, Feo, PRB 73, 052402 (2006); [S12] Cea et al., arXiv:2402.11022 (2024); [S13] Beccaria, Campostrini, Feo, PRB 76, 094410 (2007); [S19] Temme, Bravyi, Gambetta, PRL 119, 180509 (2017); [S21] Bonet-Monroig et al., PRA 98, 062339 (2018); [S23] Sandvik, AIP Conf. Proc. 1297, 135 (2010); [S28] our channel-response derivation; [S30] Wecker, Hastings, Troyer, arXiv:1507.08969 (2015). Tools: PennyLane 0.44.1, NumPy, SciPy, Matplotlib.